

Towards LLM-powered Drone Companion for Blind and Low Vision Users

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Abstract

In this work, we present an initial step towards an LLM-powered assistive drone for Blind and Low Vision (BLV) users. Through a formative study (N=9), we identified envisioned use cases and desired interaction modalities. Then, we took a participatory design approach to build a prototype, incorporating the feedback received from 3 BLV users, as well as 5 domain experts. We found that participants were receptive to the idea of an LLM-powered assistive drone and envisioned using such a system to address unmet needs.

CCS Concepts

• **Human-centered computing** → **User studies; Empirical studies in accessibility.**

Keywords

Large Language Models, Drones, Accessibility

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1 Introduction

Vision impairment affects at least 2.2 billion people worldwide¹. In recent years, the HCI community has seen a wave of assistive robots designed to support Blind and Low Vision (BLV) community in their daily lives. For example, robotic canes and robotic guide dogs have been explored to replace their non-technical counterparts [7, 10, 13]. Collectively, these systems demonstrate that assistive robots have great potential to improve the quality of life for BLV people and, importantly, the form factor of assistive robots significantly affects both the range of tasks they can perform and their overall usability.

Drones have emerged as a versatile form of assistive robots, allowing them to support BLV users in a variety of tasks, such as navigation [1, 8] and locating objects [5]. Furthermore, a recent survey found that BLV would like drones to provide assistance that expands beyond mundane day-to-day activities to leisure activities [4]. Unlike prior works where the BLV users must have a physical hold of the robot to use it, there is greater space leniency between users and the drone – for instance, a drone could be flying meters ahead of the user, which opens up new use cases and interactions.

To fully harness the potential of a versatile assistive robot and bridge the distance between users' goals and robots' capabilities, past research has highlighted that BLV users have greatly benefited from voice-based interactions [2]. Recent advances in large language models (LLMs), such as ChatGPT, have transformed this landscape by enabling robots to understand and respond in natural language, enabling natural and intuitive communication between

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¹<https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment>

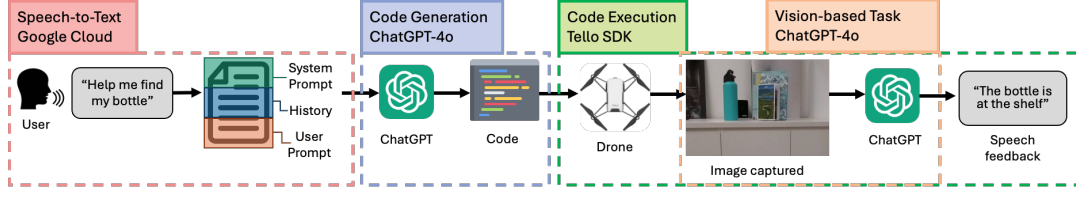


Figure 1: The overview of the software implementation of our prototype. Google Speech-to-Text transcribes the user’s command, which is then combined with the system prompt and sent to GPT-4o to generate code for the drone. The generated code is run using the Tello SDK, and any vision-based tasks are handled through GPT-4o’s multimodal features. Finally, the response is read out to the user.

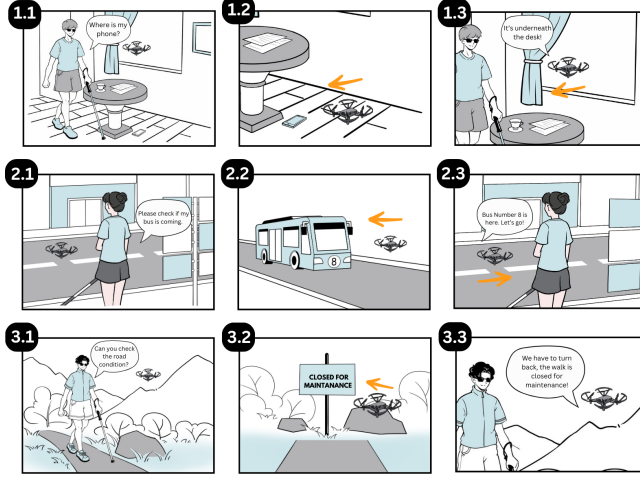


Figure 2: Illustration of use cases mentioned by the participants. They consist of using the drone to locate a mobile phone indoors, checking the bus arrivals at a stop, and assessing walking path conditions outdoors.

users and robots [6]. For instance, Padmanabha et al. have built an LLM-powered feeding robot for the elderly. In this case, the elderly only need to provide natural language commands, while the LLM would parse human intent and generate step-by-step instruction for the robot [9]. However, limited work has explored harnessing the power of LLMs for assistive robots for BLV individuals.

Drawing from and expanding upon these works, we take a step further to build an LLM-powered voice-based assistive drone for BLV people. We started with a formative study with 9 BLV people to understand potential use cases and key design considerations for an assistive drone. Then, we took a participatory design approach to build a prototype [11]. This involved a series of design iterations, incorporating the feedback received from 3 BLV people, as well as 5 domain experts. The insights gathered from these activities not only shed light on aspects to improve our prototype, but also promote understanding of the implications of introducing novel assistive technology to BLV people.

2 Identifying Requirements

Method: We started our investigation with a formative study with 9 BLV users (refer to Appendix A) to identify requirements for such assistive drone - such as the needs and challenges of BLV

users in their daily lives, as well as their envisioned use cases and interaction modalities for assistive drones. All studies in this paper were approved by the Institutional Review Board Committee of our university.

Design Implications: Building on our findings and related literature, we formalized the following design goals: 1) **Versatility:** Participants were receptive to the idea of a drone companion and recognized the distinct advantages that this form factor offers. They envisioned that the assistive drone should be able to address unmet needs for the BLV community - for instance, P6 wanted to use a drone to locate misplaced objects like a phone accidentally dropped on the floor (refer to Figure 2 for more examples). 2) **Natural Communication:** Participants preferred voice inputs, which aligns with previous work [4]. Participants also highlighted the importance of flexible voice input - for example, P1 expressed frustration with current voice assistants: “every time I say something to Siri, it doesn’t get it, then I must rephrase, which is very annoying.” 3) **Safety:** As highlighted in prior work [9], another important consideration is ensuring the drone operates in a predictable and controlled manner to build user trust and minimize physical risks.

3 Prototype Walkthrough

We used the Tello EDU drone², a lightweight quadcopter drone, to build the proof-of-concept prototype. The process begins with the user uttering a voice command, which is then transcribed and sent to the GPT-4o backend for code generation. As shown in Figure 1, the pipeline comprises the following components:

Speech-to-Text: To enable users to communicate with the drone intuitively and naturally, we deployed a Speech-to-Text module to allow voice inputs. Upon receiving the wake-up word or the button press, the audio recording is initiated; then the recorded audio is transcribed. We used Google Speech-to-Text³ transcription.

Code Generation: Following previous guidelines [9, 12], we created a specialized prompt for GPT-4o to generate Python code based on the user’s command. The prompt includes the following elements (see Figure 1, left): context by specifying the role of GPT-4o, a list of high-level functions, instructions to generate Python code, the history of the previous conversations, and the command provided by the user. To support code execution, in addition to providing a list of high-level functions in the prompt, the complete implementation of these functions can be found in the function library.

²<https://www.ryzerobotics.com/tello>

³<https://cloud.google.com/speech-to-text/docs>

Code Execution: We used the `exec()` function to dynamically execute Python code stored in a text file. To control the drone's movement and obtain its status, the system used Tello EDU SDK with the help of a wrapper library⁴. Vision-based tasks, such as object detection and text recognition, are performed by GPT-4o, which supports multimodal inputs. To summarize, GPT-4o plays a pivotal role in two critical stages: First, we use GPT-4o to generate code from users' natural utterances. Second, we leverage GPT-4o's multimodal capabilities to process visual input and generate contextually relevant responses.

4 Initial User Feedback

To iterate the prototype in a meaningful direction, we gathered feedback from 5 domain experts and 3 BLV users (refer to Appendix A) to further improve our prototype. We encouraged participants to think creatively and state any modifications or improvements they could think of after interacting with the prototype. Overall, most participants had a smooth interaction, and they offered insights regarding both the functionalities and interactions. For instance, the transcription was not always accurate (e.g., “plant” vs “clamp”). While misinterpretation is one of the most common voice assistant issues [3], we found that such errors are particularly disruptive to assistive robots, as they directly lead to task failure. Also, some BLV users suggest that it would be helpful to create separate functions for “count” and “color” to improve identification accuracy. Regarding latency time, we observed that longer prompts would increase response time significantly during extended conversations.

5 Future Work and Conclusion

We plan to iterate on our prototype by incorporating the suggestions from our participants - for instance, we plan to implement multi-threading, allowing API calls to ChatGPT to run concurrently with the drone's movements, which will minimize delays and improve user experience. We also plan to conduct a user study with additional BLV users. We plan to evaluate the effectiveness and usability of our prototype with both qualitative and quantitative data. In addition, we hope to promote understanding of the implications of introducing new assistive technology into the lives of BLV people.

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⁴<https://djitellopy.readthedocs.io/en/latest/tello/>

A Appendix A

Table 1: Focus Group Discussion Participant Information

ID	Age	Gender	Vision Level	Past Experience
P1	49	F	less than 5% of her central vision	sport enthusiast; familiar with assistive technologies
P2	49	M	fully blind since birth	familiar with assistive technologies
P3	40-50	F	partially blind with hearing loss	housewife
P4	65	F	partially blind	retired
P5	31	M	fully blind due to a disease	tech-savvy
P6	59	M	fully blind since birth (born with glaucoma)	familiar with assistive technologies
P7	40	M	legally blind due to genetic issue (can perceive light)	familiar with assistive technologies
P8	45	M	fully blind due to glaucoma	sport enthusiast; tech-savvy
P9	63	F	fully blind in one eye, and less than 5% in the other	administrative officer; familiar with assistive technologies

Table 2: Co-design Expert Information Expert

ID	Age	Gender	Past Experience
P1	30-35	M	PhD holder in HCI and haptics
P2	40-45	M	entrepreneur with 10+ years experience working with people with disabilities, including BLV users; familiar with a variety of hardware devices, including drones
P3	25-30	F	designer with extensive experience working with people with disabilities
P4	50-55	M	PhD in human-drone interaction; drone hobbyist (10+years)
P5	30-35	F	PhD in audio; have worked with blind users and drones

Table 3: Co-Design Blind Participant Information

ID	Age	Gender	Vision Abilities	Experience with Assistive Technologies
P1	32	M	Fully blind	Cane user. Used to work with drones in media and production. Tech-savvy.
P2	69	M	Partially blind due to glaucoma	Cane user. Used to work in the aviation section. Tech-savvy.
P3	20	M	Partially blind since birth	Cane user. Seldom use assistive technologies.